

Eliminating Wrong-Direction Travel in Elevator Systems

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1. Problem

In a normal elevator, passengers can only press ↑ or ↓ in the hallway. The system doesn't know where they want to go. As a result, an elevator going up may pick up a passenger who wanted to go down, carry them upward first, and only then bring them back down. This wastes time and frustrates passengers.

2. Goal

Design a smarter elevator system that knows **the passenger's destination before assigning a car**, so no one is ever sent in the wrong direction.

3. System Design

3.1 Number of Elevators

For a 10-floor office building with 200 people, the system must move about 30 people every 5 minutes during peak time. Based on standard elevator traffic guidelines, **3 elevators** meet this requirement comfortably while keeping waiting times short. The simulation therefore uses 3 elevators.

3.2 Elevator Selection (Score Function)

Instead of pressing ↑ or ↓ in the hallway, the passenger enters their **destination floor** on a keypad. The system calculates a **score** for each elevator and assigns the one with the lowest score:

Score = pick up time + drop off time + wrong-direction penalty + load penalty

The wrong-direction penalty is set very large, so an elevator going the opposite way is never chosen. This is the rule that directly solves the wrong-direction problem.

3.3 Time-Based Parking

During rush hour, idle elevators do not remain stationary. Instead, they **automatically return to the lobby**, so they are pre-positioned for the next wave of arriving passengers. This reduces waiting time during predictable peak periods.

3.4 Cancelling a Call

If a passenger presses the wrong destination, they can simply **press the same button again to cancel**. The button works as a toggle: pressing it once calls the elevator, pressing it again cancels the call. This keeps the interface simple and feels familiar, since it works the same way as a conventional elevator button.

4. Expected Results

The new system is expected to:

- **Eliminate wrong-direction travel:** no passenger will ever be carried opposite to their destination.
- **Reduce average journey time** compared to a conventional elevator, especially during rush hour.
- **Pre-position elevators** at the lobby during peak times, reducing waiting time for arriving passengers

Demo:

<https://drive.google.com/file/d/10cLxDVTBZF7e6jjiYGX1krJh4val4cdC/view?usp=sharing>